Peroxide mixture quantification

Rapid and Facile Spectrophotometric Method for the Selective Determination of Peroxymonosulfate, Peroxydisulfate and Hydrogen Peroxide in Ternary Mixtures

Source: 10.26434/chemrxiv.15000337/v1 | 13 PDF pages

All page citations count from the first PDF page, including its cover.

96-well polystyrene plate	250 uL per well	350 nm at 25 °C	0–10 s / 3 min / 8 min

Extracted scope

The three iodide-based microplate assays for PMS, hydrogen peroxide and PDS. Begins with prepared samples and standardized peroxide stocks; electrolysis and iodometric stock standardization are outside this extracted assay module.

Status: source extraction for review. Unresolved details are retained. No physical run or executable CubOS file is claimed.

Materials and starting conditions

Input	Reported conditions / unresolved details	PDF p.
Sample and matching matrix	Prepared peroxide sample; ultrapure water or the same electrolyte as the sample for blanks.	5
KI stock	Reported as 1.2 M: 19.9 g KI dissolved in 100 mL ultrapure water. The text describes solvent volume, not an explicit final-volume make-up.	5
Phosphate stock	pH 4.0. The text labels it 0.1 M but later uses 0.5 M and gives 68 g KH2PO4 in 1 L water. Concentration is unresolved; retain both reported labels.	5
Molybdate stock	Reported as 35 mM: 0.124 g (NH4)6Mo7O24·4H2O dissolved in 20 mL ultrapure water. Confirm whether the molarity is expressed per Mo equivalent.	5
Bicarbonate stock	2 M KHCO3. Its stock-preparation quantities are not given here.	5
Readout	Agilent PowerWave HT microplate reader; polystyrene 96-well plate; absorbance at 350 nm and 25 °C.	5

Ordered procedure

P01 Prepare indicator mix 1 (KI)

Combine 14 mL phosphate stock (specified here as 0.5 M, pH 4.0) with 1 mL of 1.2 M KI. The authors report 80 mM KI in mix 1. Resolve the conflicting phosphate-stock labels before preparation.

Source: PDF page(s) 5 | reported with conflict

> Source excerpt: 14 mL phosphate buffer stock (0.5 M, pH 4.0) with 1 mL KI stock (1.2 M)

P02 Prepare indicator mix 2 (KI/Mo)

Combine 13.83 mL phosphate stock (specified here as 0.5 M, pH 4.0), 667 uL of 1.2 M KI and 500 uL of the reported 35 mM molybdate stock. Reported mix concentrations: 53 mM KI and 1.2 mM molybdate.

Source: PDF page(s) 5 | reported with conflict

> Source excerpt: 667 uL KI stock (1.2 M) and 500 uL molybdate stock (35 mM)

P03 Prepare indicator mix 3 (KI/KHCO3)

Combine 12.5 mL of 1.2 M KI with 7.5 mL of 2 M KHCO3. The authors report 0.75 M each of KI and KHCO3 in mix 3.

Source: PDF page(s) 5 | reported

> Source excerpt: 12.5 mL KI stock (1.2 M) and 7.5 mL KHCO3 (2 M)

P04 Check indicator-stock background

Keep stocks refrigerated in the dark. The paper describes storage for several weeks and regular fresh preparation; it discarded and remade indicator stocks when blank absorbance at 350 nm exceeded 0.15. No precise shelf life or refrigerator temperature is given.

Source: PDF page(s) 5 | reported

> Source excerpt: absorbance of the blank at 350 nm was above 0.15

P04b Establish each branch's calibration

Use standardized PMS for KI calibration; separate PMS and H2O2 standards for the combined KI/Mo calibration; and separate PMS, H2O2 and PDS standards for total-peroxide calibration. Reported quantification ranges are 7-250 uM, 4-250 uM and 17-500 uM respectively. Individual calibration points and dilution-transfer recipes are not specified here. Verify calibration in the actual plate geometry as the authors require.

Source: PDF page(s) 5, 7 | reported

> Source excerpt: quantification range of 7 - 250 uM

P05 Load the PMS assay and matrix blank

Add 100 uL sample to each assay well, with at least two wells for the same sample. In a separate blank well, add 100 uL water or the matching electrolyte. These are separate wells, not additions to one well.

Source: PDF page(s) 5 | reported

> Source excerpt: 100 uL of a sample containing various peroxides were placed in a well

P06 Start and read the PMS assay

Add 150 uL mix 1 to each sample and blank well. Measure absorbance at 350 nm, 25 °C immediately after addition. Results specify a 0-10 second window; the authors describe approximately 3-5 seconds between addition and their first measurement. Subtract the matrix-blank reading and use the KI-assay calibration.

Source: PDF page(s) 5, 6, 8 | reported

> Source excerpt: 150 uL of mix 1

P07 Load and start the combined PMS + H2O2 assay

In fresh wells, add 100 uL sample or matching matrix blank, then 150 uL mix 2. Include replicate measurements as described for the PMS assay. This branch measures the combined PMS and H2O2 response.

Source: PDF page(s) 5 | reported

> Source excerpt: 150 uL of mix 2 were added to all the prepared wells

P08 Read the combined PMS + H2O2 assay

Begin reading at 350 nm immediately after mix 2 addition. Record for 3 minutes at 10-second intervals at 25 °C. Use the baseline-corrected mean between 2 and 3 minutes with the KI/Mo calibration.

Source: PDF page(s) 5 | reported

> Source excerpt: for 3 minutes in 10 s intervals at 25 °C

P09 Load and start the total-peroxide assay

In another set of wells, mix 50 uL sample with 200 uL mix 3. For the corresponding background, combine 50 uL matching matrix with 200 uL mix 3. Replicate determination is recommended.

Source: PDF page(s) 5 | reported

> Source excerpt: 50 uL of sample were mixed with 200 uL of mix 3

P10 Read the total-peroxide assay

At 25 °C, measure absorbance at 350 nm for 8 minutes at 10-second intervals, starting immediately after indicator addition. Baseline-correct and use the mean between 5 and 8 minutes with the total-peroxide calibration.

Source: PDF page(s) 5 | reported

> Source excerpt: mean absorbance between 5 and 8 minutes

P11 Calculate the three analyte concentrations

PMS comes from the KI branch. H2O2 = concentration from KI/Mo minus PMS from KI. PDS = total concentration from KI/KHCO3 minus the combined concentration from KI/Mo. Apply each branch's calibration to concentration before subtraction; raw absorbances are not interchangeable.

Source: PDF page(s) 5 | reported

> Source excerpt: [H2O2] = [PMS + H2O2]KI/Mo − [PMS]KI

Derived arithmetic – separate from source instructions

Final well volumes	100 + 150 = 250 uL for KI and KI/Mo; 50 + 200 = 250 uL for KI/KHCO3. Arithmetic from reported transfers.
Sample dilution in the readout	2.5-fold for the first two branches and 5-fold for the third. Arithmetic; verify the calibration concentration basis before comparing or subtracting concentrations.
Indicator dilution in the wells	The written mix recipes and additions imply 48 mM KI in branch 1, approximately 32 mM KI / 0.72 mM molybdate in branch 2, and 0.60 M KI / KHCO3 in branch 3. Arithmetic from stock labels; the Results use 80 mM, 53 mM / 1.2 mM and 0.75 M labels. Resolve stock-versus-final concentration conventions before using the method.

Details to resolve before execution

Conflicting phosphate-stock concentration

The stock paragraph says 0.1 M, while both mix recipes specify 0.5 M. The 68 g/L recipe is also consistent with approximately 0.5 M KH2PO4. The extraction preserves this conflict instead of silently choosing a value.

Source: PDF page(s) 5

Stock-versus-final concentration labels

The recipe adds indicator stock to sample, diluting it. Results label conditions with the same concentrations as those indicator stocks. Clarify whether these are stock or final-well labels; do not change either recipe silently.

Source: PDF page(s) 5, 6, 8

Fast timing and unspecified mixing

PMS is measured within 0-10 seconds; the reported initial read was after about 3-5 seconds. Mixing method, speed and duration are absent. Dispensing/read latency must preserve the stated kinetic window.

Source: PDF page(s) 5, 6, 8

Calibration and stock verification need supporting detail

The paper refers to SI for titration and Table S1. Those separate supplements are not included with this selected main-article PDF. Exact calibration preparation and dilution accounting need review. The authors explicitly say calibration depends on the container/path length.

Source: PDF page(s) 5, 6

Molybdate concentration basis

The salt contains seven Mo atoms. Preserve the stated mass, solvent volume and "35 mM" label; confirm the intended molarity basis rather than assuming salt molarity.

Source: PDF page(s) 5

Original file: chemrxiv.15000337_v1.pdf

Source SHA-256: 5e1fe338c46e59ce70be779efd42f40ec41087fcb986558b47e6427c1c3389bd

Chloride transport in vesicles

High-throughput and Automated Anion Transport Assays

Source: 10.26434/chemrxiv-2024-kxf0x | 10 PDF pages

All page citations count from the first PDF page, including its cover.

Manual 96-well assay	303 uL conditional total	369 / 503 nm fluorescence	5 min after each salt spike

Extracted scope

The manual 96-well lucigenin chloride/nitrate antiport assay. Begins with prepared lucigenin-loaded POPC vesicles and transporter stocks. The later 384-well automated workflows are separate methods and are not merged into this extraction.

Status: source extraction for review. Unresolved details are retained. No physical run or executable CubOS file is claimed.

Materials and starting conditions

Input	Reported conditions / unresolved details	PDF p.
Prepared POPC vesicles	Body text: 2 mM POPC; internal buffer 100 uM lucigenin, 200 mM NaNO3 and 5 mM sodium phosphate, pH 7.2. Figure 1 instead labels internal lucigenin 1 mM. This discrepancy is unresolved.	4
External buffer	200 mM NaNO3 with 5 mM sodium phosphate, pH 7.2.	4
Transporter stocks	A concentration series of transporter in DMSO, described as 100X stocks. The individual stock concentrations and dilution-transfer recipe are not specified in the core paragraph.	4
NaCl spike	200 mM NaCl with 5 mM sodium phosphate, pH 7.2.	4
Controls and reader	DMSO-only vesicles for 0% influx; an unspecified highly potent transporter for 100% influx. Fluorescence plate reader with autoinjector.	4

Ordered procedure

C01 Load the prepared vesicles

Dispense vesicle suspension into the 96-well plate. The PDF literally prints "150 L per well"; the u prefix is missing on the rendered page, not just in text extraction. A proposed interpretation is 150 uL, but this quantity remains unresolved in the structured record.

Source: PDF page(s) 4 | unit unresolved

> Source excerpt: 150 L per well, 2 mM POPC

C02 Add transporter and mix

Add 3 uL of the appropriate 100X transporter stock in DMSO to each assay well. Mix using a multichannel pipette. Exact mixing volume, cycle count and speed are not stated.

Source: PDF page(s) 4 | reported

> Source excerpt: 3 uL of 100X stock solution

C03 Include reference conditions

Include vesicles treated with DMSO only as the 0% chloride-influx reference and vesicles treated with a highly potent transporter as the 100% reference. Keep DMSO concentration consistent across wells, as described in the preceding assay discussion. The reference-transporter identity and dose must be resolved.

Source: PDF page(s) 3, 4 | reported

> Source excerpt: vesicles treated with DMSO only as a reference for 0% chloride influx

C04 Start chloride influx

Use the plate-reader autoinjector to add 150 uL of the NaCl-spike solution to each well. Start each well's endpoint timing from its own salt addition.

Source: PDF page(s) 4 | reported

> Source excerpt: 150 uL per well, 200 mM NaCl

C05 Read the five-minute endpoint

Measure fluorescence five minutes after NaCl addition: excitation 369 nm, emission 503 nm. Assay temperature, reader gain and bandwidths are not stated in this paragraph.

Source: PDF page(s) 4 | reported

> Source excerpt: excitation = 369 nm, emission = 503 nm

C06 Convert and normalize the signal

Use the Stern-Volmer calibration to convert fluorescence to chloride concentration, then calibrate against the 0% and 100% influx references before fitting the final Hill plots. The paper points to supplementary Figure S28 and Equation S2 for these transformations.

Source: PDF page(s) 4 | reported

> Source excerpt: creation of a Stern-Volmer plot (Figure S28)

C07 Keep batch repeats distinct from well repeats

The reported manual-method validation used three repeated Hill plots from three different vesicle batches. Do not convert that statement into an invented requirement for three identical wells per concentration.

Source: PDF page(s) 6 | reported

> Source excerpt: 3 repeated Hill plots were performed in 3 different batches of vesicles

Derived arithmetic - separate from source instructions

Conditional final volume	If the missing unit in C01 is confirmed as uL: 150 + 3 + 150 = 303 uL. Conditional arithmetic, not a reported 300 uL total.
Conditional DMSO fraction	3 / 303 × 100 = 0.990% v/v if the transporter stock is in neat DMSO and C01 is 150 uL. Conditional calculation. Keep identical across the concentration series.

Details to resolve before execution

Printed loading-volume unit is incomplete

The visual PDF says "150 L". The proposed 150 uL interpretation is plausible for a microplate but is explicitly not treated as a resolved source value.

Source: PDF page(s) 4

Lucigenin concentration conflicts

The procedure says 100 uM internal lucigenin; Figure 1 labels 1 mM. This tenfold difference must be reconciled from the supplementary method or authors.

Source: PDF page(s) 4

Upstream preparation and calibration are external

Vesicle manufacture, the transporter dilution series, reference transporter identity/dose, Stern-Volmer coefficients and Equation S2 are not fully specified in the selected main-PDF procedure. The separate SI is not in the 100-file ZIP.

Source: PDF page(s) 4

Plate capacity and unspecified parameters

The conditional total is 303 uL, not 300 uL. Plate working volume, mixing parameters, temperature, exact well mapping and wells per concentration require decisions; none are invented here.

Source: PDF page(s) 4, 6

Vesicle age affects the signal

The paper warns that leakage can reach 20% within two hours and later discusses use within 2-3 hours. A hard validity window should be reviewed for the chosen vesicle batch.

Source: PDF page(s) 4, 7

Original file: chemrxiv-2024-kxf0x.pdf

Source SHA-256: e94e10f4d393d01cab68226f8fb4e2d3a9cb81c7848c89a87fcddc8c059827f9

Nucleoside kinase activity

Semi-automated high-throughput substrate screening assay for nucleoside kinases

Source: 10.26434/chemrxiv-2021-k0w7q | 13 PDF pages

All page citations count from the first PDF page, including its cover.

150 uL enzyme reaction	100 uL luminescence well	19 h at 37 °C + heat stop	10 min dark; 1 s read

Extracted scope

The standard nucleoside-kinase reaction followed by the Kinase-Glo Max residual-ATP readout, using the human deoxycytidine kinase branch as the concrete example. Enzyme production, HPLC validation and optimized conditions for other enzymes remain separate.

Status: source extraction for review. Unresolved details are retained. No physical run or executable CubOS file is claimed.

Materials and starting conditions

Input	Reported conditions / unresolved details	PDF p.
NK reaction buffer	70 mM Tris, pH 7.6; 10 mM MgCl2; 5 mM DTT.	11
ATP and nucleoside	Final reaction concentrations 0.4 mM ATP and 0.33 mM nucleoside. General stocks are described as 100, 10 or 1 mM in deionized water, without assigning the exact stock to every reaction.	10–11
Enzyme	Standard reaction contains 0.0002 U enzyme per 150 uL. Human deoxycytidine kinase (HsdCK/NK13) is an available branch; stock activity in U/uL is not given.	10–11
Detection reagent	Kinase-Glo Max reagent (Promega), used at 10 uL per readout well. Reconstitution/preparation follows an external reagent method and is not specified in this PDF.	11
Labware and reader	96-well PCR reaction plate; 96-well white detection plate; Synergy Mx multimode reader with 1-second luminescence integration and extended dynamic range.	11

Ordered procedure

K01 Prepare reaction mixtures

Prepare 150 uL reactions in NK buffer containing final 0.4 mM ATP, 0.33 mM nucleoside and 0.0002 U enzyme. The source gives final composition, not exact stock-transfer volumes or order; do not manufacture those missing pipetting instructions.

Source: PDF page(s) 11 | reported

> Source excerpt: total volume of 150 uL consisting of 0.4 mM ATP, 0.33 mM nucleoside and 0.0002 U enzyme

K02 Prepare matched reaction controls

Prepare a negative control without enzyme, a substrate control without ATP, and a basal-activity control without nucleoside substrate. Incubate them under the same conditions as the reactions. The exact replacement volumes for omitted stocks are not stated.

Source: PDF page(s) 11 | reported

> Source excerpt: negative control without enzyme

K03 Incubate the enzyme reaction

Incubate at 37 °C for 19 hours. In the plate implementation, reactions are sealed with EASYseal film and arched auto-sealing lids preheated to 50 °C. Kinase reactions were performed in duplicate.

Source: PDF page(s) 11, 12 | reported

> Source excerpt: incubated at 37°C for 19 h

K04 Stop the reaction by heating

Heat to 75 °C for 10 minutes. The PCR-cycler implementation specifies lid heating at 85 °C. The paper does not give a cooling/equilibration step or temperature before adding the luminescence reagent; this remains an explicit handoff gap.

Source: PDF page(s) 11 | reported

> Source excerpt: 75°C for 10 min with lid heat set to 85°C

K05 Prepare standards for the ATP readout

Prepare ATP standards in NK buffer spanning 0-500 uM. The exact standard points are not listed in the core procedure. Standards are assayed twice; samples are assayed three times. Keep this detection replication distinct from the duplicate enzyme reactions.

Source: PDF page(s) 11, 12 | reported

> Source excerpt: standards ranging from 0 to 500 uM of ATP

K06 Dilute the sample in the detection well

Combine 10 uL stopped reaction sample or standard with 80 uL deionized water in a white 96-well assay plate. Figure 1C explicitly places water dilution before reagent addition for standards. Applying that same order to reaction samples is an interpretation to confirm; page 11 specifies their amounts but not the water/reagent addition order.

Source: PDF page(s) 5, 11 | partly inferred order

> Source excerpt: 10 uL phosphate donor standard was mixed with 80 uL deionized water before adding 10 uL of Kinase-Glo reagent

K07 Add the luminescence reagent

Add 10 uL Kinase-Glo reagent to each detection well. The resulting final assay volume is 100 uL. Mixing parameters are not reported.

Source: PDF page(s) 11 | reported

> Source excerpt: 10 uL sample/standard and 10 uL Kinase-Glo reagent

K08 Incubate and read

Incubate in the dark for 10 minutes, then measure luminescence for 1 second with extended dynamic range. The temperature of this 10-minute detection incubation is not stated.

Source: PDF page(s) 11 | reported

> Source excerpt: after 10 min incubation in the dark

K09 Apply the reported control calculation

As printed: consumed ATP (%) = 100 − 100 × (R + B + S)/(N + S). R is mean reaction luminescence, N is mean no-enzyme-control luminescence, S is substrate-control signal, and B is the difference between negative-control and basal-activity signals. Product formation (%) = consumed ATP (%) × 1.2, restricted to 0–100%. These are the paper's equations, not calculations on experimental data.

Source: PDF page(s) 11 | reported

> Source excerpt: multiplying the consumed ATP with the factor 1.2

Derived arithmetic – separate from source instructions

Detection volume	10 uL sample + 80 uL water + 10 uL reagent = 100 uL. Arithmetic matches the explicit reported total.
Detection dilution	The sample/standard is 10% of the readout volume: a tenfold dilution. Use the same handling for standards; do not apply dilution correction twice.
Reaction amounts	150 uL × 0.4 mM = 60 nmol ATP; 150 uL × 0.33 mM = 49.5 nmol nucleoside. Derived amounts; exact stock transfer volumes still require stock concentrations/activity.

Details to resolve before execution

Reaction setup lacks component volumes

The final volume/concentrations are explicit, but per-stock transfer volumes, selected stock concentrations, enzyme stock activity and dispensing order are not. These fields remain null in JSON.

Source: PDF page(s) 10, 11

Cooling and readout temperature are absent

No post-75 °C cooling/equilibration procedure is stated, and the temperature of the 10-minute dark readout incubation is unspecified.

Source: PDF page(s) 11

Calibration points and reagent preparation

The 0–500 uM ATP range is reported, but exact standard points, dilution-transfer recipes and Kinase-Glo preparation are not supplied in the core method.

Source: PDF page(s) 11

Source analysis needs expert review

The ATP-consumption equation and B definition are transcribed exactly as reported. Control arithmetic and the approximate 1.2 substrate ratio should be reviewed before interpreting assay data. The paper recommends HPLC confirmation or an enzyme-dose repeat for low conversions.

Source: PDF page(s) 6, 11

Original file: chemrxiv-2021-k0w7q.pdf

Source SHA-256: 6a2a607ea5b671b73e5e93afcbaae686458f6c5689dc219cd38e6fb25e40774e

Extraction record

Direct PDF text review plus visual inspection of the key Methods pages. Page references count from the first PDF page, including the ChemRxiv cover. No external protocol was substituted.

All 100 titles/first pages indexed. Candidate method sections inspected. These three replace the earlier Bradford, DPPH and beer selections.

The 28 source excerpts were checked against the text extracted from the cited PDF pages. Key Methods pages were visually inspected to verify numbers and units, including a missing volume prefix and conflicting concentration labels.

Reported values, conditional interpretations, derived arithmetic and unresolved details are distinguished. A matching excerpt establishes traceability; it does not establish scientific validity.

The review pack includes this PDF, an HTML review with source-page images, three individual protocol JSON files, a combined JSON file, the exact three original PDFs and verification.json.

No original download or previously created protocol was overwritten. The original 100-paper ZIP and extracted collection remain available.

For the HTML version, open index.html from the same review-pack folder.